

Caoliwen Wang

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Research Interests

Rooted in computer graphics, scientific computing, and generative AI, my research develops **physically grounded, design-ready, and scalable generative models** for physical world simulation, video generation, 3D shape generation, and scientific discovery.

Education

University of British Columbia

Sep 2025 - Present

Ph.D. in Computer Science

Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China

Sep 2021 - Jun 2025

B.S. in Computational Mathematics

Advisor: Prof. Ligang Liu

Research Experience

Physics-grounded Video Generation

2025 - Present

- **WorldParticle**: a unified world model that represents diverse physical phenomena as Lagrangian particles and learns their dynamics with a single Transformer-based backbone, enabling physics-grounded video generation and 3D dynamic generation.
- **Physical Action-Conditioned Video Generation** (in progress): a real-time action-conditioned video generation framework that integrates a physics-grounded neural simulator, enabling controllable video generation with physically plausible dynamics.

Physically Plausible 3D Generation

2025

- **Gauss–Seidel Projection**: a method that enforces physical validity (non-penetration, structural plausibility) on generated 3D protein configurations and improves the inference efficiency of protein structure prediction foundation models.

Design-Driven 3D Shape Generation

2024 - 2025

- **Neural Shadow Art**: an implicit neural 3D representation that facilitates geometry generation from multi-view binary shadow targets, with angular adjustment and volume/geometry optimization.
- **Shape from Semantics**: a 3D generation framework for creating semantically consistent shapes whose multi-view renderings align with user-specified prompts, extending the inverse-modeling framework to a semantics-conditioned 3D generation problem.

Publications & Preprints

- [1] **Caoliwen Wang**^{*}, Minghao Guo^{*†}, Siyuan Chen^{*}, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen[†]. **WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer**. *Preprint, Under Review at SIGGRAPH Asia, 2026*.
- [2] Siyuan Chen^{*}, Minghao Guo^{*†}, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen[†]. **Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection**. *International Conference on Learning Representations (ICLR), 2026*.
- [3] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang[†]. **Shape from Semantics: 3D Shape Generation from Multi-View Semantics**. *Preprint, Under Review at TVCG, 2025*.
- [4] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang[†]. **Neural Shadow Art**. *Pacific Graphics (Conference Track), 2025*.

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)